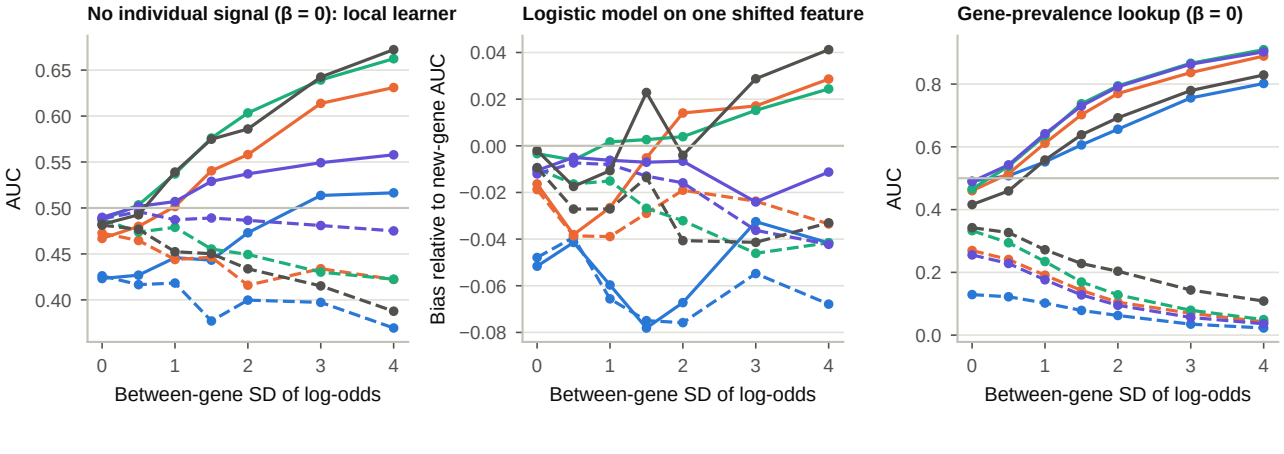
No individual signal (β = 0): local learner
AUC
0.65
0.60
0.55
0.50
0.45
0.40
0
1
2
3
4
Between-gene SD of log-odds
Logistic model on one shifted feature
Bias relative to new-gene AUC
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
0
1
2
3
4
Between-gene SD of log-odds
Gene-prevalence lookup (β = 0)
AUC
0.8
0.6
0.4
0.2
0.0
0
1
2
3
4
Between-gene SD of log-odds
12 genes × 2
12 genes × 5
12 genes × 10
48 genes × 5
cohort-matched (18 genes)
variant-level CV
gene-level CV